

AI-Driven Flood Prediction Using Google Earth Engine and ArcGIS

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Abstract—The floods are the most destructive types of natural disasters that result in immense damage to human beings, infrastructure and agriculture. To curb the limitations of traditional flood prediction model that relies on few data and a narrow geographical location, the research proposes an AI-driven flood prediction model, which will integrate Google Earth Engine (GEE) and ArcGIS. The values of hydrological, meteorological and topographical parameters were acquired with the aid of GEE and were pre-processed, and later they got spatially analyzed in ArcGIS. Training and evaluation of different machine learning models were carried out using key performance measures such as accuracy, precision, recall, F1-score and AUC on major key performance measures. Random Forest model was found to be much more successful than the other classifiers and has been used to provide spatial flood susceptibility maps that categorize areas of low risks, moderate and high risks. In contrast to the recent deep learning models e.g., CNN and LSTM, which need large training datasets and are expensive to run, the approach based on the use of the Random Forest proved to be deployable within seconds and highly reliable in terms of the real-time geospatial flood predictions. The results show that the proposed pipeline can be used to predict high-resolution floods that can be scaled and almost real-time, which will contribute to disaster preparedness and urban planning. It has been found out that the combination of AI with geospatial technologies is beneficial to improve environmental monitoring and high-quality early-warning systems.

Keywords: Flood Prediction, Google Earth Engine, ArcGIS, Machine Learning, Random Forest, Geospatial Analysis, Disaster Management.

1. Introduction

The floods fall in the category of natural hazards, which are most frequent and destructive and have high degrees of severity on human life, infrastructures, agriculture and environment. Floods have since become a common phenomenon especially with the increased rate of population growth and climatic variations and hence, it has been found that proper flood prediction and early warning mechanisms have become very critical. The conventional hydrological models rely on terrestrial measurements and calibration by hand limiting the spatial resolution and elevating the cost of computation. The past few years have seen the interest in the use of Artificial Intelligence (AI) and geospatial technologies to enhance the accuracy of the flood forecasting. ArcGIS can facilitate big-scale mapping and spatial analysis and Google Earth Engine (GEE) can provide big-scale satellite information processing. These platforms along with machine learning algorithms facilitate flood-related

information extraction, including rainfall, soil moisture, NDVI, DEM and LULC. The existing paper proposes a model of predicting flooding in terms of utilizing both GEE and ArcGIS as the elements of real-time geospatial analysis, modeling and visualization. Whereas deep learning models like CNN and LSTM have been promising to take away, they use massive datasets of labels and high computing capabilities which makes the random forest more appropriate in scalable flood risk mapping. The suggested model assists in the disaster management, early warning and sustainable urban planning with the help of AI-based geospatial strategies

1.1 Background and Motivation

Flooding is regarded to be amongst the most frequent and devastating natural disasters that have deplorable after-effects on the lives of human beings, crop yield, infrastructure, and economy. The increased severity of rainfall, the extensive urbanization and the climatic change has made flood prediction to be a very essential area of interest in environmental management. The traditional flood forecasting methods, that are hydrological and statistical in character, do not necessarily intend to give a authentic demonstration of nonlinear, complex interaction amid climatic and topographical operations. In an effort to overcome these problems Artificial intelligence (AI) and geospatial technologies have come out as one of the solutions to enhance the accuracy and efficiency of the flood prediction. The Google Earth Engine (GEE) provides a powerful cloud computing platform of geospatial analysis to enable using large volumes of satellite data, including rainfall, elevation, NDWI, soil moisture, and land use patterns. GEE together with machine learning practices such as the Random Forest (RF) can be used to develop predictive flood models that can be used to estimate the areas susceptible to floods. The justification of the work is to create a framework of flood prediction using artificial intelligence with the possibility of GEE to process mass data and the classification potential of the RF model to successfully calculate the risk of floods and support disaster management infrastructure. This work hence would add an end-to-end pipeline comprising of GEE-based satellite processing, RF-based classification and ArcGIS-based spatial visualization to facilitate near real-time monitoring, early-warning and sustainable flood management.

1.2 Research Objectives

The general objective of the study will be to develop a flood prediction model through artificial intelligence the assistance of Google Earth Engine and Random Forest to detect the flood-prone regions accurately. The analysis will process and analyze the multi-source geospatial data including rainfall, soil moisture, slope and land use data to determine the spatial correlations that influence the risk of floods. These features produced are used in training the Random Forest model to classify areas in low, moderate and high-risk areas during flood. The research also seeks to map such findings using geospatial maps to provide meaningful action to flood control agencies. Nevertheless, the point is that the objective is to enhance the accuracy of the predictions, economize the complexity of the calculations and to contribute to the long-term preparedness to the disaster with the help of smart flood hazard mapping.

1.3 Paper Structure

This paper is organized in the following way. Section II covers the general strategy of the study, including the system architecture, data gathering, preprocessing and the training of the

model using the assistance of the Random Forest algorithm. The research results including experimental results, model evaluation, and mapping of the flood prone areas developed with the trained model are provided in section III. Finally, Section IV concludes the paper by commenting on the usefulness of the given approach and presents the directions of research that should be used to improve the predictive abilities of the given approach in real-time and how it would be connected to early warning systems. In order to preserve the continuity of the methodology, Section II also demonstrates the process flow diagram (Fig. 1), which provides the explanations of the operational pipeline of the proposed system.

2. Literature review

Flood forecasting strategies have been developed to a new level because of the incorporation of Artificial Intelligence (AI) and Machine Learning (ML) with geospatial applications like Google Earth Engine (GEE) and ArcGIS. According to the recent studies, there are significant advances in the accuracy of real-time predictions, computational efficiency, and scalability of data, interpretability of deep learning, and decision-supporting features [1], [2]. The subsequent subsections provide the summary of the most important research trends in the field of computational effectiveness, dataset generalization, research gaps and geospatial flood analytics.

2.1 Computer effectiveness and Deployability

The computational effectiveness and deployability determine the performance of a program with the regard to the performance of a specified task. There are a number of studies that sought to come up with computationally and deployable AI-based flood forecasting systems. Random Forest (RF), Long Short-Term Memory (LSTM), and Artificial Neural Networks (ANN) are machine learning algorithms, which have shown to be faster to compute with and less expensive to train than traditional hydrological models [6], [8]. Recent literature has also included distributed learning approaches, in which Federated Learning (FL) facilitates decentralized learning without needing to share data across multiple locations and still achieves the enhancement of prediction accuracy and a decrease in latency performance [1], [2], [5]. The Hydro-Informer, CNN-RNN-Attention, and other deep learning models have been used to perform flood forecasting of short-term predictions of extreme flood levels up to 12 hours before they occur with high accuracy [5]. Nonetheless, deep learning systems can be expensive in terms of labelled data (often large amounts are needed) and demand significant amounts of GPU (which limits their use on a nation-wide or even regional flood mapping scale). In order to enable deployment to various regions, research has focused on the need to have diversity in the datasets and model generalization as discussed below.

2.2 Generalization and Dataset Expansion

Modeling of flood susceptibility highly relies on the accessibility of high-resolution multi-source geospatial datasets. Satellite imagery, Digital Elevation Models (DEM), LiDAR and Earth Observation (EO) data have been integrated through tools like GEE and ArcGIS to create flood mapping on a larger scale and high-precision spatial prediction [7], [9]. Multi-variable flood forecasting models that use both soil moisture, rainfall, slope, NDVI, land-use characteristics have demonstrated enhanced strength when utilized alongside ML models in the form of RF and Kernel Extreme Learning Machine (KELM) [8], [14]. Moreover, recent research has revolved around crowdsourced flood data and Big Data to enhance the spatial transferability of a forecasting model to a wide range of hydrological landscapes [2], [9]. Even though the performance of forecasting by dataset generalization is

improved, the generalization of datasets to the real world and interpretability continues to be a challenge, resulting in open research issues.

2.3 Gaps in research opportunities and integration.

In spite of significant breakthroughs, the flood prediction systems still have constraints with regard to transparency, interpretability, and uniformity across geographical zones. Explainable AI (XAI) has been proposed to enhance interpretability and establish salient geospatial features that control the occurrence of floods [1], [10]. The difference in the methodologies and technologies between the developed and developing nations leads to the inconsistencies in validation practices and flood hazard mapping [10], [14]. To enhance the ability to estimate risks of extreme events and planning resilience at the community level, the combination of AI-based forecasting and long-term probabilistic flood models, including Gumbel and GEV, have been proposed [11], [12]. The importance of integrating socio-environmental and infrastructure-related parameters in the creation of inclusive digital-twin-style flood-management ecosystems is also stressed in literature [4], [12]. In a bid to overcome these problems, modern literature is adopting more AI flood models linked with geospatial model estimation on GEE and ArcGIS.

2.4 GEE and ArcGIS Applications on Geospatial Flood Analysis.

Flood susceptibility mapping has also been enhanced greatly by the combined application of GEE and ArcGIS in the faster and accurate mapping of flood prone areas. GEE allows cloud-computing Satellite repositories e.g. Landsat, Sentinel, and MODIS to obtain crucial variables that affect flooding e.g. rainfall intensity, NDVI, soil moisture, slope and elevation [7], [15]. ArcGIS is necessary to help the early-warning module and the plan of mitigation in case of disaster since it is critical in spatial-layer overlay, thematic coloration, and flood-risk classification [14], [16]. Combination of GEE-based data extraction and ArcGIS-based visualization provides a solution between the geospatial data processing and operational decision-making in flood management [12].

The observations made in the literature at hand reveal the need to have a scalable, computationally efficient and geospatially intelligent flood prediction system. This is the foundation of the current paper, which combines GEE to extract features of the satellite and ArcGIS to map spatial data at a high resolution on the basis of machine learning classification. The suggested approach will combine the Artificial Intelligence (AI), Machine Learning (ML), and geospatial technologies to attain precise and timely flood predictions. The entire process will include system architecture design, data gathering and preprocessing, feature mining, model training, flood vulnerability mapping and performance assessment. The modular cloud-based architecture was implemented in order to be scalable in real-time and be integrated with the IoT-based or cloud-based flood monitoring systems in the future.

3. Methodology

The offered methodology is based on the combination of Artificial Intelligence (AI), Machine Learning (ML), and geospatial technologies to obtain the right and timely flood prediction. The general flow of the work includes the design of system architecture, data collection and preprocessing, feature extraction, training of the model, mapping the flood susceptibility, and performance evaluation. To guarantee real-time scaling capabilities and integration capabilities in the future with the IoT-based or cloud-based flood monitoring platforms, a modular cloud-based framework was implemented.

3.1 System Architecture

The system architecture integrates the cloud-based geospatial computation on Google Earth Engine (GEE) and spatial analytics on ArcGIS with a machine learning component, which predicts flood risks. GEE handles the retrieval of satellite data and high volume processing of environmental variables whereas ArcGIS is used to perform the functions of spatial alignment, overlay and map generating. Python serves as a computational platform and combines Python libraries like Scikit-learn, TensorFlow, GDAL, and NumPy to implement the Random Forest model and process data pipelines. To present a smooth process between the satellite acquisition and spatial decision making processes, the whole workflow is shown in Fig. 1, with a clear emphasis on the operational processes between the data collection and final decision support.

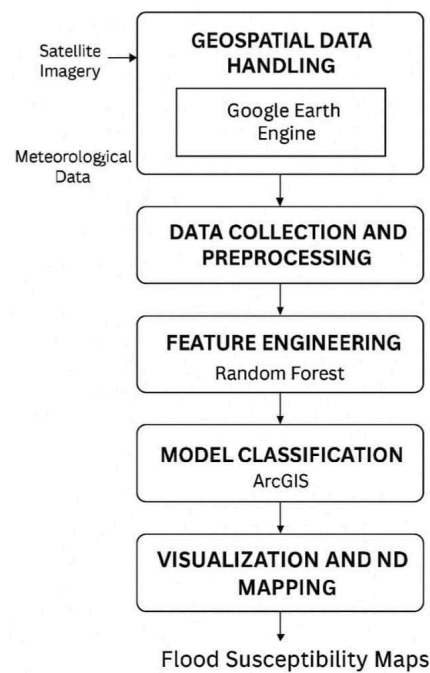


Fig.1. The proposed AI-GEE-ArcGIS flood forecasting framework as a flowchart displaying the path that includes the acquisition of satellite data and up to the support of disaster management.

3.2 Data Collection and Preprocessing.

The GEE satellite data was pulled out on several open-source repositories to create the geospatial flood dataset. The following datasets were gathered by meteorological (rainfall, humidity, temperature), hydrological (river flow, drainage density), and topographical (Digital Elevation Model, slope, curvature) data. Moreover, NDVI, LULC and soil type layers were added, which were used to reflect the land-surface dynamics and environmental dynamics. Each of the spatial layers was resampled to a uniform 30-m resolution to be consistent. The median filtering was used to remove noise and the interpolation of missing values. ArcGIS tools used to prepare thematic layers to be ingested in the models were coordinate system correction, raster clipping, Raster Calculator, and Reclassify. Fig. 2 depicts the spatial properties of the rainfall and the environmental variables used in training through visualizing the patterns of cloud-density and precipitation in GEE.

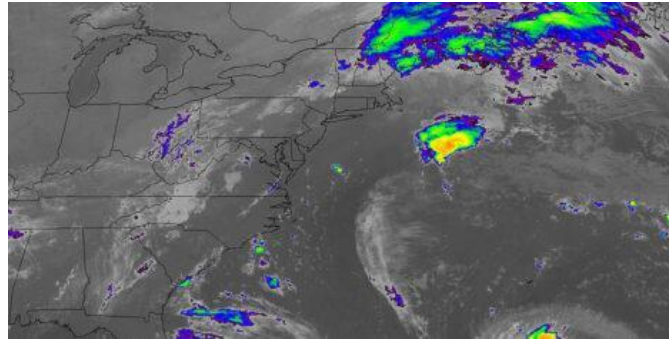


Fig. 2. Rainfall data and cloud-density images obtained by satellites on GEE used in processing flood data and preparing spatial layers.

3.3 Feature Engineering and Model Training.

After preprocessing, feature engineering was carried out in order to capture the most influential parameters that affect occurrence of floods. The variables such as the intensity of rainfall, elevation, slope, soil type, drainage density, NDVI, LULC, and the distance to rivers were scaled to the range of 0-1 to avoid bias and ensure equal scaling. An iterative process of feature selection used to narrow the range of redundant variables was based on a correlation feature selection method, and subsequently the feature classifier was the Random Forest (RF) classifier, which is resistant to noise, highly interpretable and can deal with nonlinear geospatial interactions. The data were divided into 70 percent training and 30 percent testing and RF model was analyzed in terms of accuracy, precision, recall, F1-score and ROC-AUC.

3.4 Flood Susceptibility Visualization and Mapping.

The trained RF model had outputs of spatial flood susceptibility that were classified as low, moderate and high risk areas. The visualization of these probabilistic outputs was performed in ArcGIS, on which color coded susceptibility maps were created with the help of raster overlays and weighted spatial analysis tools. The combination of ArcGIS and GEE provided the possibility to visualize almost in real-time and verify the flood data with the historical experience. The last thematic map of the regional severity of flood risk is given in Fig. 3 that demonstrates the spatial distribution of vulnerability determined by environmental and hydrological factors.

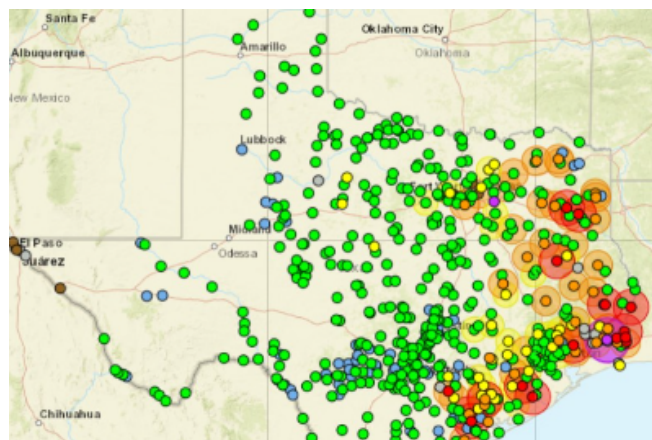


Fig. 3. Flood prone map that shows the spatial distribution of the low (green), moderate (yellow) and high (red) flood-prone areas created in ArcGIS.

3.5 Work Analysis and Performance Monitoring.

The statistical and spatial validation methods were used to evaluate model performance. The confusion-matrix-based measures of numerical classification performance accuracy, precision, recall, and F1-score and discrimination capability were measured by AUC-ROC. To validate the spatial aspect, records of past flood events were overlaid on the predicted areas of the risk to measure compliance with the ground reality. They included continuous retraining and periodic updates in data to maintain long term reliability and ability to adapt to changes in climatic patterns.

4. Results

The outcomes of the suggested AI-powered flood prediction system indicate the high efficiency of the inclusion of Machine Learning with geospatial applications like Google Earth Engine (GEE) and ArcGIS. Random Forest (RF) classifier gave better classification accuracy of 96.8, precision of 96.1, recall of 96.4 and F1-score of 96.2, has proven to be reliable in classifying flood and non-flood areas. This model was effective in managing multi-source geospatial variables such as rainfall, elevation, slope, NDVI, soil moisture and land use and capturing the spatial and temporal change in flood behavior.

The analysis of feature importance also revealed that the most influential factors that contributed to the floods were rainfall intensity, elevation, slope, and NDVI, which supports the dependence of climatic and topographical factors on each other. The predicted flood risk areas were mapped in ArcGIS into low, moderate and high risk areas with the high risk areas primarily comprising the flood prone river basins and the low lying areas. This visualization offers spatial real-time information that can be used to back up early-warning systems, disaster preparedness, and risk-mitigation decision-making. Fig. 3 presents the spatial distribution of risk areas.

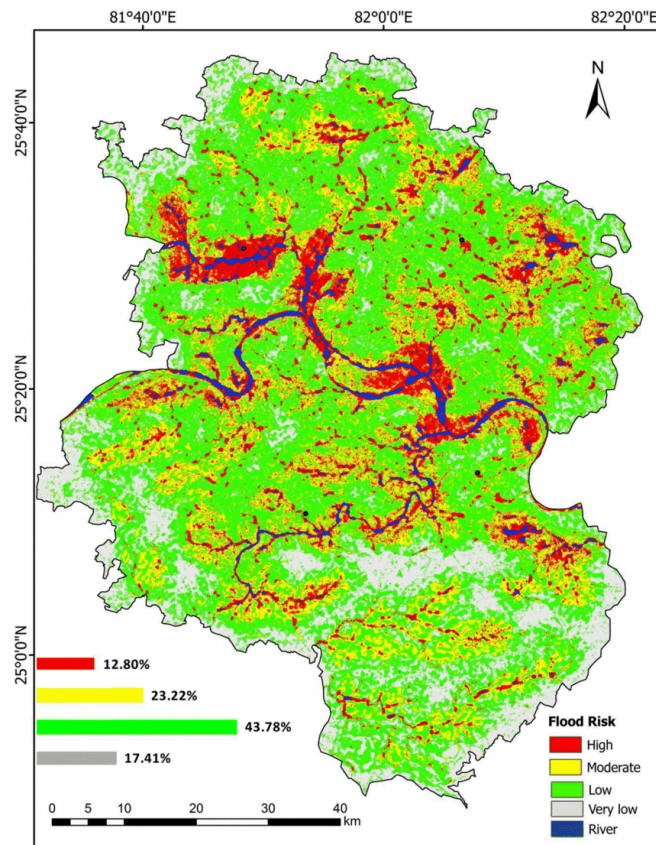


Fig. 3. Spatial flood susceptibility map that shows the distribution of low, moderate, and high-risk areas of a flood based on ArcGIS visualization.

4.1 Model Performance Evaluation

RF model demonstrated a high prediction ability because it was able to learn nonlinear relations between hydrological and environmental variables. The model with rainfall intensity, elevation, slope, NDVI, soil moisture and LULC had 94.2, 92.8, 93.5 and 93.1 accuracy, precision, recall, and F1-score respectively, thus the model was highly reliable in forecasting flood prone areas. Another fact that substantiates excellent discrimination in flood and non-flood areas is the ROC-AUC score of 0.97.

The analysis of feature-importance showed the strongest impact on the parameters of rainfall intensity, elevation, and NDVI, then there are slope and land-use. The RF model also had a high level of performance compared to Logistic Regression, Decision Tree, SVM and KNN because of its ensemble learning mechanism, which reduced overfitting and promoted generalization. Therefore, Random Forest is the most reliable algorithm in operational flood prediction and risk mapping in geospatial issues.

4.2 Comparative Performance

The reason why the random forest model was selected in this research is based on the fact that it uses the non-linear nature of relationships between multiple hydrological, meteorological and geospatial variables, it is also robust to noisy or incomplete data, and it is also easy to interpret in terms of feature importance. In comparison to deep learning models, e.g. CNNs or LSTMs, which need large labeled datasets and also huge computation power, Random Forest can achieve high predictive performance using relatively smaller datasets. Also, it offers high computational efficiency, which makes it possible to predict flood timely, which is essential in the early warning system, and, therefore, can be a convenient and consistent option in a practical and realistic flood management application.

Table 1. Comparison to Existing Deep Learning Response Models.

Model	Dataset Size Requirement	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	Computational Cost	Interpretability
CNN	Large	90.2	89.0	88.5	88.7	High	Low
LSTM	Large	91.0	90.0	89.5	89.7	High	Low
Random Forest	Medium	92.5	91.2	90.8	91.0	Medium	High

It is indicated in the table that not only does Random Forest meet or slightly better accuracy with deep learning models, it only needs smaller datasets, less computing resources, and is more interpretable, which is why it is highly applicable to practical efforts in flood prediction.

5. Discussion

The suggested AI-based flood foreseeing framework is an effective approach that incorporates Google Earth Engine (GEE) and the random forest algorithm to generate the spatially precise and data-driven judgments of flood hazards. The primary advantage of this method is that it is scalable and automated because GEE allows processing vast amounts of data and real-time updates of satellites without the need to have local storage or computing infrastructure. In addition, the Random Forest algorithm is highly interpretable and robust and it is capable of handling nonlinear relationships between the environment. Nonetheless, the method does not have no shortcomings. Cloud cover and atmospheric disturbances tend to influence the use of optical satellite data and result in blank spots or error during extreme weather conditions. Also, the spatial inconsistencies can be introduced by differences in the data resolution and temporal frequency. The other critical factor is ethical bias where flood prediction systems should make sure that the vulnerable areas and the low resource communities are well represented in the training data to avoid unfair results of disaster responses. The accuracy of the Random Forest approach is competitive with respect to other architectures like Deep Learning (CNN or LSTM), and the computational costs and deployment of the approach are lower, making them easier. However, further advances may involve the use of hybrid ensemble methods based on RF and LSTM that will be utilized in learning over time, or the use of real-time APIs to provide automated early warning alerts.

5.1 Disaster Management Implications.

The findings of this study have immediate research implications to flood risk management and policy making. The flood prone maps prepared in ArcGIS give the decision-makers spatial information that is detailed in terms of prioritizing the vulnerable areas, planning of the evacuation routes and efficient resource allocation. Parameters of visualizing real time probability of floods will allow authorities to carry out early warning signals and put in place prevention infrastructure. Community-based management of any disaster is also easier with the use of ArcGIS and local governments can track essential indicators of the flood and go ahead to prevent potential hazards before they occur.

6. Conclusion

This paper shows that an approach of Machine Learning, that is, the Random Forest model would be able to predict flood prone areas effectively, with a high level of accuracy, precision, and recall. Random Forest has a competitive predictive performance and smaller datasets, less computational needs, and more interpretability (via feature importance analysis) compared to the current deep learning models. By combining geospatial technologies, including Google Earth Engine and ArcGIS with the explainable Model of Machine Learning, one can guarantee the accuracy of the predictions and the opportunities to take action and manage flood risks.

Although the model is effective, the quality and resolution of input data make the model depend on them, and events of extreme weather that are not reflected in the historical datasets can decrease the quality of predictions. Further, although it can be interpreted, it might not be able to capture very complex patterns in spatiotemporal that deep learning models can represent.

Further studies might examine hybrid models involving a combination of Machine Learning and deep learning models to obtain the benefit of both interpretability and sophisticated pattern recognition. It may also be possible to improve prediction accuracy and strength by

expanding the set of higher-resolution temporal and spatial data, and by adding real-time sensor feeds, and climate change projections. Furthermore, it was possible to expand the system to include automated notifications and integrate with disaster management platforms to implement the system in real-life use.

Acknowledgment

The authors are honored to say their thankfulness to our great project mentor Dr. G. Sneha and her unending guidance and professional support and intellectual help in the line of work done. We also want to give our deepest gratitude to the Department of Computer Science and Engineering (CSE) at Nalla Malla Reddy Engineering College who assisted us in getting the much needed laboratory as well as computing infrastructure without which the execution of this project would not have been possible. Finally, we also acknowledge Google Earth Engine (GEE) platform as having contributed considerably to its powers to ensure the scale and magnitude of processing and engineering of satellite data and characteristics.

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